

Large-scale replication and heterogeneity in Parkinson disease genetic loci

Manu Sharma, PhD
John P.A. Ioannidis, MD
Jan O. Aasly, MD
Grazia Annesi, MD
Alexis Brice, MD
Christine Van Broeckhoven, PhD
Lars Bertram, MD
Maria Bozi, MD
David Crosiers, MD
Carl Clarke, MD
Maurizio Facheris, MD
Matthew Farrer, PhD
Gaetan Garraux, MD
Suzana Gispert, MD
Georg Auburger, MD
Carlos Vilarinho-Güell, PhD
Georgios M. Hadjigeorgiou, MD
Andrew A. Hicks, PhD
Nobutaka Hattori, MD, PhD
Beom Jeon, MD
Suzanne Lesage, PhD
Christina M. Lill, MD
Juei-Jueng Lin, MD
Timothy Lynch, MD
Peter Lichtner, PhD
Anthony E. Lang, MD
Vincent Mok, MD
Barbara Jasinska-Myga, MD
George D. Mellick, PhD
Karen E. Morrison, MD
Grzegorz Opala, MD
Peter P. Pramstaller, MD
Irene Pichler, PhD
Sung Sup Park, MD
Aldo Quattrone, MD
Ekaterina Rogaeva, PhD
Owen A. Ross, PhD
Leonidas Stefanis, MD
Joanne D. Stockton, MD
Wataru Satake, MD, PhD
Peter A. Silburn, MD
Jessie Theuns, PhD
Eng-King Tan, MD
Tatsushi Toda, MD, PhD
Hiroyuki Tomiyama, MD, PhD
Ryan J. Uitti, MD
Karin Wirdefeldt, MD, PhD
Zbigniew Wszolek, MD
Georgia Xiomerisiou, MD
Kuo-Chu Yuch, MD
Yi Zhao, MD
Thomas Gasser, MD
Demetrius Maraganore, MD
Rejko Krüger, MD
On behalf of the GEO-PD Consortium

Correspondence & reprint requests to Dr. Sharma: manu.sharma@uni-tuebingen.de

Editorial, page 619

Supplemental data at www.neurology.org

Supplemental Data



ABSTRACT

Objective: Eleven genetic loci have reached genome-wide significance in a recent meta-analysis of genome-wide association studies in Parkinson disease (PD) based on populations of Caucasian descent. The extent to which these genetic effects are consistent across different populations is unknown.

Methods: Investigators from the Genetic Epidemiology of Parkinson's Disease Consortium were invited to participate in the study. A total of 11 SNPs were genotyped in 8,750 cases and 8,955 controls. Fixed as well as random effects models were used to provide the summary risk estimates for these variants. We evaluated between-study heterogeneity and heterogeneity between populations of different ancestry.

Results: In the overall analysis, single nucleotide polymorphisms (SNPs) in 9 loci showed significant associations with protective per-allele odds ratios of 0.78–0.87 (*LAMP3*, *BST1*, and *MAPT*) and susceptibility per-allele odds ratios of 1.14–1.43 (*STK39*, *GAK*, *SNCA*, *LRRK2*, *SYT11*, and *HIP1R*). For 5 of the 9 replicated SNPs there was nominally significant between-site heterogeneity in the effect sizes (I^2 estimates ranged from 39% to 48%). Subgroup analysis by ethnicity showed significantly stronger effects for the *BST1* (rs11724635) in Asian vs Caucasian populations and similar effects for *SNCA*, *LRRK2*, *LAMP3*, *HIP1R*, and *STK39* in Asian and Caucasian populations, while *MAPT* rs2942168 and *SYT11* rs34372695 were monomorphic in the Asian population, highlighting the role of population-specific heterogeneity in PD.

Conclusion: Our study allows insight to understand the distribution of newly identified genetic factors contributing to PD and shows that large-scale evaluation in diverse populations is important to understand the role of population-specific heterogeneity. *Neurology*® 2012;79:659–667

GLOSSARY

CI = confidence interval; GEO-PD = Genetic Epidemiology of Parkinson's Disease; GWAS = genome-wide association studies; HWE = Hardy-Weinberg equilibrium; MALDI-TOF = matrix-assisted laser desorption/ionization time-of-flight; MSA = multiple system atrophy; OR = odds ratio; PD = Parkinson disease; SNP = single nucleotide polymorphism.

Genome-wide association studies (GWAS) have provided tangible gains in understanding the genetic architecture of complex diseases,^{1,2} including Parkinson disease (PD).³ Several GWAS have been conducted in PD in Caucasian populations and only 1 in the Asian population.^{3–11} Consistent and reproducible association signals were confirmed in α -synuclein (*SNCA*), leucine-rich repeat kinase 2 (*LRRK2*), and microtubule-associated protein tau (*MAPT*), thus underscoring the importance of these 3 genes in the pathophysiology of the common sporadic forms of PD.^{3–10,12} In addition to that, different studies have provided some evidence for an association for *BST1*, *GAK*, and *HLA-DRB5* with PD.^{6–9,13}

A recently published GWAS meta-analysis in PD increased the number of identified PD genetic loci to 11.¹⁴ This study reported significant between-study heterogeneity for some of the 11 genetic loci¹⁴ even though data were restricted to Caucasian descent populations.

It is important to establish whether the 11 genetic loci that have been postulated to be associated with PD are replicated when tested with direct genotyping in a larger spectrum of diverse populations. The consistency or lack thereof of the genetic effects of these genetic

Author affiliations are provided at the end of the article.

Coinvestigators are listed on the *Neurology*® Web site at www.neurology.org.

Go to *Neurology.org* for full disclosures. Disclosures deemed relevant by the authors, if any, are provided at the end of this article.

variants across different populations may help to determine whether they represent genuine loci for PD susceptibility and whether they can be used for risk prediction across these diverse populations.¹⁵ To gain further insight into genetic factors contributing to PD across different populations and define the implications of between-population heterogeneity, we performed a large-scale replication study within the GEO-PD consortium.

METHODS Consortium. Investigators from the Genetic Epidemiology of PD (GEO-PD) Consortium were invited to participate in this study. A total of 21 sites representing 19 countries from 4 continents agreed to contribute DNA samples and clinical data for a total of 17,705 individuals (8,750 cases and 8,955 controls). Healthy individuals matched for age and gender served as controls. They underwent neurologic examination and were excluded from the study whenever there was clinical evidence for any extrapyramidal disorder.

Genotyping. We selected 1 SNP per each gene locus, exactly as they were proposed by the recently published GWAS meta-analysis.¹⁴ Genotyping was performed by a central genotyping core (Department of Human Genetics, Helmholtz Zentrum, Munich). Each site provided 100–200 ng of DNA to the laboratory core. In total 11 SNPs located in and around the genes encoding *SYT11*, *ACMSD*, *STK39*, *LAMP3*, *GAK*, *BST1*, *SNCA*, *HLA-DRB5*, *LRRK2*, *HIP1R*, and *MAPT* were genotyped. The genotyping core was blinded to case-control status of each site. Genotyping was performed using a matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) mass spectrometry on a MassArray system (Sequenom, San Diego, CA). Cleaned extension products were analyzed by a mass spectrometer (Bruker Daltonik, USA) and peaks were identified using the MassArray Typer 4.0.2.5 software (Sequenom). Assays were designed by the AssayDesigner software 4.0 (Sequenom) with the default parameters for the iPLEX Gold chemistry and the Human Genotyping Tools ProxSNP and PreXTEND (Sequenom). All variants were genotyped in 1 multiplex assay. An experienced investigator blinded to case or control status of the samples visually checked genotype clustering. The average call rate of the variants was >97%.

In order to further enrich the samples of Asian ancestry populations, we also included GWAS data from a Japanese population (988 cases, 2,521 controls).⁶ We used r^2 threshold of 0.8–1.0 to select proxy SNPs from the Japanese GWAS. Using this threshold, we were able to capture only 3 SNPs from *BST1*, *SNCA*, and *LRRK2* genes.

Standard protocol approvals, registrations, and patient consents. The local Ethics Committee approved the study. All participants signed an informed consent.

Analysis. An exact test was used to assess whether the genotype distributions for each SNP deviated from Hardy-Weinberg equilibrium (HWE) among controls; each site was tested separately and deviation from HWE was considered significant at <0.01. We excluded data from sites where the missing rate was >5%. For our analysis, we adhered to the same allele coding as in the previous GWAS meta-analysis.¹⁴

For consistency effect estimates based on minor vs major allele contrast were computed. We used an additive model adjusted for age and gender to obtain effect estimates. Results were then synthesized using fixed and random effects models. Fixed effect models assume that the genetic effect is the same in populations from different sites and that observed differences are due to chance alone. For associations showing between-study heterogeneity, fixed effect estimates yield narrower confidence intervals (CIs) and smaller p values as compared to random effects models, which incorporate between-study heterogeneity.^{16–18} Fixed effects analysis tests the null hypothesis of no association in all studied populations that are analyzed. Routinely, this assumption is used in GWAS settings to increase the power of meta-analysis to detect associations that may exist in some (at least 1) population. However, in presence of heterogeneity the effects may differ substantially in different populations and not all populations may show a genetic effect for the variant of interest. Random effects models allow the genetic effects might be different due to genuine heterogeneity that may exist across different sites. Random effects calculations take into account the estimated between-study heterogeneity. We used the inverse variance method for fixed effects models. Cochran Q test of homogeneity and the I^2 metric were used to evaluate the between-site heterogeneity. The Q statistics follows χ^2 -based distribution with $k - 1$ degrees of freedom (k = number of studies). I^2 is estimated by the ratio $(Q - df)/Q$, where df is degrees of freedom. The I^2 metric ranges from 0% to 100% and measures the proportion of variability that is beyond chance. Typically estimates of $I^2 < 25\%$ are considered to reflect little or no heterogeneity, 25%–50% moderate heterogeneity, 50%–75% large heterogeneity, and >75% very large heterogeneity. It should be acknowledged that I^2 can have large uncertainty in its estimation especially for variants with low minor allele frequency. Therefore, we also estimated the 95% CI of I^2 .¹⁷

The overall main analysis considered all sites and populations irrespective of ancestry. Then, we separately analyzed Caucasian and Asian sites and we compared the genetic effects in these 2 major ancestry groups.

The SNPs evaluated in the recently published GWAS meta-analysis are common with minor allele frequencies varying from 13% to 46%,¹⁴ except for SNP, rs34372695 (*SYT11*) where the minor allele frequency is 2%. Therefore, based on minor allele frequency and effect estimates obtained in the GWAS meta-analysis,¹⁴ power calculations showed that our study would have at least 99% power to detect an allele-based odds ratio (OR) of 1.2 for minor allele frequencies of 10% or higher for $\alpha = 0.05$. Based on genome-wide significance level ($\alpha = 5 \times 10^{-8}$), our study would have 43% power to detect an allele-based OR of 1.2 for minor allele frequency of 10%, but it would be 99% for same minor allele frequency and on OR of 1.4. Power would be only 69% for a minor allele frequency of 2% and OR of 1.2, but it would be 99% for the same minor allele frequency of 2% and an OR of 1.5.

Meta-analyses were performed using STATA 9.0 (Stata Corp., College Station, TX) and Review Manager 4.2.7. p Values are 2-tailed.

RESULTS Characteristics of sites and overall database. Twenty-one sites contributed a total of 8,750 cases and 8,955 controls. Characteristics of all participating sites are shown in table 1. Most sites contributed participants of Caucasian ancestry ($n = 16$); 5 sites (counting also the GWAS performed in the Jap-

Table 1 Description of datasets contributed by each study site

Site	Country	No.	Case	Control	Male (%)	Female (%)	Mean AAO	Mean age at study	Diagnostic criteria
Annesi	Italy	394	197	197	204 (51.7)	190 (48.2)	61.5	63.7	UKPDBB
Brice ^a	France	505	272	233	302 (59.8)	203 (40.1)	47.6	57.8	UKPDBB
Bozi	Greece	222	114	108	107 (48.1)	115 (51.8)	69.9	74.5	UKPDBB
Wszolek	US	1,518	692	826	794 (52.3)	724 (47.6)	64.4	71.7	UKPDBB
Garraux	Belgium	82	68	14	45 (54.8)	37 (45.1)	62.1	69.6	UKPDBB
Hadjigeorgiou	Greece	714	357	357	379 (53.0)	335 (46.9)	63.4	63.7	UKPDBB
Jeon	Korea	749	408	341	314 (41.9)	435 (58.0)	57.6		UKPDBB
Opala	Poland	629	352	277	340 (54.0)	288 (45.7)	50.2	68.1	UKPDBB
Lynch	Ireland	740	368	372	340 (45.9)	400 (54.0)	50.5	70.7	UKPDBB
Lin	Taiwan	320	160	160	160 (50)	160 (50)	62.0	70.8	UKPDBB
Facheris	Italy	181	114	67	86 (47.5)	95 (52.4)	63.0		UKPDBB
Maraganore	US	1,024	801	223	600 (58.5)	361 (35.3)	59	74.7	Bower
Mellick	Australia	2,024	1,012	1,012	1,042 (51.4)	981 (48.4)	59	72.2	Bower
Morrison ^a	England	1,120	766	354	606 (54.1)	514 (45.8)	66.1		UKPDBB
Mok	China	436	260	176	264 (60.5)	170 (38.9)		63.5	UKPDBB
Aasly	Norway	1,278	656	622	721 (56.4)	557 (43.5)	58.8	72.9	UKPDBB
Wirdefeldt	Sweden	299	83	216	147 (49.1)	152 (50.8)	65.8	71.4	Gelb
Van Broeckhoven	Belgium	1,010	501	509	500 (49.5)	509 (50.3)	60.5	66.3	Pals/Gelb
Rogaeva	Canada	560	387	173	303 (54.1)	257 (45.8)	49.7	64.2	UKPDBB
Tan	Singapore	391	194	197	244 (62.4)	147 (37.5)	59.7	54.0	UKPDBB
Toda	Japan	3,509	988	2,521	1,844 (52.6)	1,665 (47.4)	58.7	66.0	UKPDBB
Total		17,705	8,750	8,955			59.5	67.6	

Abbreviations: AAO = age at onset; GWAS = genome-wide association studies; UKPDBB = UK Parkinson's Disease Brain Bank.

^a Also included in the previously published GWAS.^{7,8}

anese population⁶) included participants of Asian ancestry. We excluded 1 site with 114 cases and 67 controls from the analysis due to a plate layout error. The median age at onset was 59 years and median age at examination was 67 years.

We observed that for one site, effect estimates for all SNPs were “inverse” as compared to other Caucasian sites. Allele flipping for one particular site in the same Caucasian descent might reflect error in sampling ascertainment and is unlikely to reflect genuine effects.¹⁹ This site (n = 181) was therefore excluded from further analyses. Overall, genotype call rates were >97%. The genotype distribution for each SNP in the controls of each site showed no departure from HWE, except for rs6599388 (*GAK*) in samples from 4 Asian sites. We therefore excluded this SNP (rs6599388) from analyses in the Asian population.

Overall data synthesis. We observed consistent and reproducible associations for *SNCA*, *LRRK2*, *MAPT*, *BST1*, *GAK*, *STK39*, *SYT11*, *LAMP3*, and *HIP1R* loci but not for *ACMSD* (rs10928513) or *HLA-DRB5* (rs3129882) where the per-allele OR was very close to the null (1.02 and 0.95, respectively) and

statistically nonsignificant (table 2). Thus we provide unequivocal support for the involvement of these newly identified genetic loci in the pathogenesis of PD.

Summary effect estimates were generally comparable with the previous GWAS meta-analysis results (table 2), although effect estimates in this study were stronger for *STK39* and somewhat weaker for *LRRK2* compared to the previous GWAS meta-analysis.¹⁴ Exclusion of 1,625 samples that overlap with the previously published GWAS did not change any of the estimates (table e-1 on the *Neurology*[®] Web site at www.neurology.org). The protective per-allele OR ranged from 0.78 to 0.87 (*LAMP3*, *BST1*, and *MAPT*) and the susceptibility per-allele OR ranged from 1.14 to 1.43 (*STK39*, *GAK*, *SNCA*, *LRRK2*, *SYT11*, and *HIP1R*). Cochran Q statistics were nominally significant for *STK39*, *LAMP3*, *BST1*, and *SNCA* with *I*² estimates ranging from 39% to 48%. The heterogeneity reflected primarily differences in the magnitude of the effect sizes across different sites, while the direction of the effect was consistent in all sites, with rare exceptions.

Table 2 Overall analysis irrespective of ethnicity and influence of between-study heterogeneity

Gene	SNP	Q test p value	I ² (95% CI)	Odds ratio (95% CI) by random effects	Fixed effects p value	Random effects p value	IPDGC study odds ratio (fixed effects p values)
<i>ACMSD</i>	rs10928513	0.71	0 (0-54)	1.02 (0.96-1.08)	0.479	0.479	1.07 (0.003)
<i>STK39</i>	rs2102808	0.02	46 (0-68)	1.21 (1.08-1.35)	0.0001	0.001	1.12 (0.0016)
<i>LAMP3</i>	rs11711441	0.04	39 (0-64)	0.85 (0.77-0.94)	3.01×10^{-5}	0.002	0.87 (6.92×10^{-5})
<i>GAK</i>	rs6599388	0.06	39 (0-64)	1.19 (1.10-1.28)	3×10^{-6}	0.001	1.14 (7.46×10^{-8})
<i>HLA-DRB5</i>	rs3129882	0.23	18 (0-53)	0.95 (0.90-1.01)	0.12	0.15	0.80 (9.3×10^{-8})
<i>BST1</i>	rs11724635	0.02	43 (0-65)	0.87 (0.83-0.91)	2.01×10^{-6}	0.00001	0.87 (2.43×10^{-9})
<i>SNCA</i>	rs356219	0.02	48 (0-69)	1.30 (1.21-1.40)	4.23×10^{-23}	4.23×10^{-23}	1.27 (4.23×10^{-23})
<i>SYT11</i>	rs34372695	0.38	6 (0-50)	1.43 (1.15-1.78)	0.001	0.001	1.44 (1.18×10^{-6})
<i>LRRK2</i>	rs1491942	0.75	0 (0-50)	1.14 (1.07-1.21)	1.06×10^{-8}	1.06×10^{-8}	1.30 (1.06×10^{-8})
<i>HIP1R</i>	rs10847864	0.90	0 (0-44)	1.15 (1.09-1.21)	9.06×10^{-7}	9.06×10^{-7}	1.13 (9.06×10^{-7})
<i>MAPT</i>	rs2942168	0.14	29 (0-62)	0.78 (0.71-0.85)	1.37×10^{-13}	1.37×10^{-13}	0.80 (1.37×10^{-13})

Abbreviations: CI = confidence interval; IPDGC = International Parkinson disease Genomics Consortium; SNP = single nucleotide polymorphism.

Analysis including only Caucasian sites. Restricting the analysis to Caucasian sites only resulted in per-allele ORs that ranged from 0.78 to 0.90 for the 3 replicated protective loci (*BST1*, *LAMP3*, and *MAPT*) and from 1.14 to 1.43 for the 6 replicated susceptibility loci (*STK39*, *GAK*, *SNCA*, *LRRK2*, *SYT11*, and *HIP1R*), while *ACMSD* and *HLA-DRB5* still had no significant effect (table 3 and figure e-1).

Summary effect estimates were generally comparable to those of the previous GWAS meta-analysis,¹⁴ except for modest differences in *STK39* and *LRRK2* effect sizes, as noted above also for the overall analysis. There was nominally significant heterogeneity only for *SNCA* and *LAMP3* (*I*² estimates 51% and 46%, respectively), but this reflected primarily differences in the magnitude of the effect size estimates rather than direction of effects across sites (figure e-1).

Analysis including only Asian sites. In the Asian series, not only the *SYT11* SNP, but also the *ACMSD* and *MAPT* SNPs were monomorphic (table e-2). Summary effect estimates for the remaining SNPs are shown in table 4. We again observed consistent nominally significant evidence of association for all loci except for *STK39* (which still had an effect size estimate consistent with what was seen in the overall analysis) and *HLA-DRB5* (which had a point estimate very close to the null), which still had an effect size estimate consistent with what was seen in the overall analysis. Results were generally consistent across sites, with the exception of *STK39* that showed very large heterogeneity (*I*² = 73%) (figure e-2).

Comparison of effect size estimates. Five gene loci (*HIP1R*, *LAMP3*, *LRRK2*, *SNCA*, and *STK39*)

Table 3 Caucasian specific effect estimates and influence of between-study heterogeneity

Gene	SNP	Q test p value	I ² (95% CI)	Odds ratio (95% CI) by random effects	Fixed effects p value	Random effects p value
<i>ACMSD</i>	rs10928513	0.71	0 (0-54)	1.02 (0.96-1.08)	0.479	0.479
<i>STK39</i>	rs2102808	0.10	37 (0-65)	1.21 (1.08-1.35)	0.000	0.001
<i>LAMP3</i>	rs11711441	0.02	46 (0-70)	0.86 (0.76-0.97)	0.001	0.025
<i>GAK</i>	rs6599388	0.27	17 (0-57)	1.14 (1.06-1.23)	1.01×10^{-4}	0.001
<i>HLA-DRB5</i>	rs3129882	0.10	32 (0-63)	0.95 (0.88-1.02)	0.11	0.16
<i>BST1</i>	rs11724635	0.14	29 (0-62)	0.90 (0.85-0.95)	0.001	0.003
<i>SNCA</i>	rs356219	0.01	51 (0-72)	1.30 (1.19-1.42)	4.23×10^{-23}	4.23×10^{-23}
<i>SYT11</i>	rs34372695	0.38	6 (0-50)	1.43 (1.15-1.78)	0.001	0.001
<i>LRRK2</i>	rs1491942	0.80	0 (0-47)	1.15 (1.07-1.23)	1.06×10^{-8}	1.06×10^{-8}
<i>HIP1R</i>	rs10847864	0.90	0 (0-47)	1.15 (1.08-1.22)	9.06×10^{-7}	9.06×10^{-7}
<i>MAPT</i>	rs2942168	0.14	29 (0-62)	0.78 (0.71-0.85)	1.37×10^{-13}	1.37×10^{-13}

Abbreviations: CI = confidence interval; SNP = single nucleotide polymorphism.

Table 4 Asian specific effects and influence of between-study heterogeneity

Gene ^a	SNP	Q test p value	I ² (95% CI)	Odds ratio (95% CI) by random effects	Fixed effects p value	Random effects p value
<i>STK39</i>	rs2102808	0.01	73 (0–88)	1.14 (0.85–1.52)	0.28	0.37
<i>LAMP3</i>	rs11711441	0.33	12 (0–72)	0.81 (0.67–0.97)	0.01	0.03
<i>BST1</i>	rs11724635	0.07	53 (0–81)	0.74 (0.68–0.81)	1.2 × 10 ^{−6}	0.001
<i>HLA-DRB5</i>	rs3129882	0.81	0 (0–68)	0.98 (0.85–1.13)	0.85	0.85
<i>SNCA</i>	rs356219	0.49	0 (0–68)	1.24 (1.08–1.43)	0.002	0.002
<i>LRRK2</i>	rs1491942	0.31	16 (0–73)	1.13 (1.02–1.24)	0.005	0.01
<i>HIP1R</i>	rs10847864	0.36	6 (0–70)	1.14 (0.99–1.32)	0.05	0.06

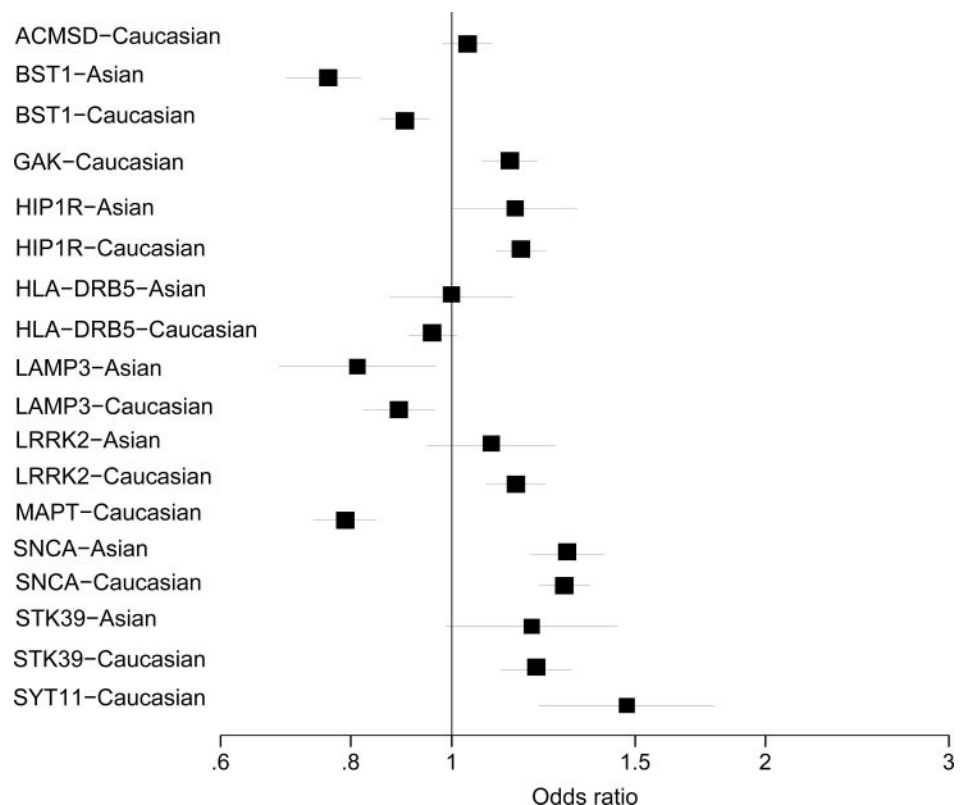
Abbreviations: CI = confidence interval; HWE = Hardy-Weinberg equilibrium; SNP = single nucleotide polymorphism.

^a ACMSD, SYT11, MAPT SNPs were monomorphic; GAK SNP showed departure from HWE and thus is not included in the table.

where both Caucasians and Asian populations were represented showed no difference in effect size estimates that were different beyond chance (figure 1). Conversely for *BST1*, the effects were different beyond chance for Asian and Caucasian populations with stronger genetic effects in the former.

DISCUSSION We performed a large-scale evaluation to assess the role of recently discovered genetic risk variants in the pathogenesis of PD in different

populations. Our study confirms 9 of the 11 postulated susceptibility SNPs for PD. The confirmed SNPs include the previously well documented *LRRK2*, *SNCA*, *BST1*, *GAK*, and *MAPT* associations,^{3,6–8,10,20} and 4 of the 5 more recently proposed associations in *STK39*, *SYT11*, *LAMP3*, and *HIP1R* SNPs that might act as risk factors for PD. Conversely, we were unable to confirm the association between PD risk and SNPs rs10928513, rs3129882 in *ACMSD* and *HLA-DRB5*, respec-

Figure 1 Forest plot showing the comparison of effect of each single nucleotide polymorphism (SNP) in Caucasian and Asian population

Boxes indicate the summary effect estimate. SNPs in 3 loci (MAPT, SYT11, and ACMSD) were monomorphic in the Asian population and thus are not included in the graph. GAK SNP showed deviation from Hardy-Weinberg equilibrium in Asian series and thus excluded from the graph.

tively. Three SNPs were monomorphic in the Asian datasets.

The recently proposed *ACMSD* locus is likely to have represented a spurious association. The OR is very close to the null and 95% CI also excludes an OR larger than 1.08. Of note, even in GWAS meta-analysis, the OR estimate was only 1.07 in the replication phase, as compared with 1.38 in the discovery phase. Our results are in agreement with a recently published study.²¹ This finding suggests that large-scale replication with direct genotyping is useful even for SNPs that pass conventional genome-wide significance thresholds. It is possible that with the use of the more extended imputation platforms using the 1,000 Genomes Project, the number of comparisons made is larger than what was done in the past with more limited imputation platforms and this may thus require more stringent levels of significance to claim genome-wide significance.

As observed in a recently published study,⁹ the frequency of the *HLA-DRB5* SNP, rs3129882, varies considerably even within seemingly homogenous Caucasian populations; the frequency of risk allele is low in subjects of Northern European descent as compared to subjects from Southern European descent. Therefore, the observed lack of association for the *HLA-DRB5*-specific SNP rs3129882 should be interpreted with caution. Moreover, directionality as well as the magnitude of effect estimates obtained in our study for the *HLA-DRB5* locus specific SNP are comparable with previously published studies.^{9,22}

Our study shed light on the role of heterogeneity in PD genetics. Detection of heterogeneity could provide new insight to understand the genetic architecture of the disease.^{17,23,24} A number of factors can be attributed to the observed heterogeneity. First, clinically overlapping pathologies may lead to heterogeneity. Indeed the presence of distinct subgroups of patients during early clinical stages of PD could contribute to clinical heterogeneity.²⁵ Therefore, it is worth it to consider that genetic variants may exert different pathologic processes that eventually lead to complex clinical phenotypes.²⁶ For example, it has been shown that multiplications of *SNCA* gene lead to clinical phenotype, which clinically overlap with multiple system atrophy (MSA).²⁷ Moreover, SNPs in the 3'UTR of *SNCA* were shown to be associated with PD as well as MSA.^{3,6–8,28} The most significant SNPs in both diseases clustered around 3'UTR of *SNCA*; SNP rs11931074, that was significantly associated with MSA in contrast to rs356219 in the PD meta-analysis.^{14,28} The r^2 between these 2 markers was only 0.16 in the Caucasian population. These 2 distinct signals in different yet overlapping pathologies therefore might reflect one cause of genetic heterogeneity. Second, heterogeneity might reflect that

different tagging polymorphisms were used in previously published GWAS. Of note, a recently published GWAS from United Kingdom, France, and Netherlands provided weak (as they did not surpass genome-wide significant threshold) yet consistent association signals for the *BST1* and *GAK* gene.^{7,8,13} This probably reflects that the investigated markers were not causal variants but in linkage disequilibrium with a potential causal variant across different studies.

Our study helps to understand the role of population-specific heterogeneity in PD risk loci. Some risk variants exist only in populations of specific ancestry. For example, it is known that an ancient inversion (~900 kb) in the *MAPT* region occurred, which led to the formation of 2 nonrecombining haplotypes, H1 and H2.²⁹ The H2 haplotype is absent in East Asian populations as has been shown also by a recently published Asian GWAS that revealed no association for the *MAPT* locus.⁶ This is in contrast to previously published candidate gene studies and GWAS in Caucasian populations that have shown consistent association with the *MAPT* locus.^{3,7,8,13}

Some other variants may have a different magnitude of effect in populations of different ancestry. In our data, this is well exemplified by *BST1*, where the OR was significantly larger in populations of Asian than those of Caucasian ancestry. Conversely, we found that associations in 5 loci (*SNCA*, *LRRK2*, *LAMP3*, *HIP1R*, and *STK39*) had similar effects in these 2 ancestries.

The consistency or diversity of effect sizes in identified associations may reflect different patterns of linkage disequilibrium in these loci in diverse populations. It may also reflect differences in the susceptibility to develop PD and perhaps also differences in age at onset in different ancestry groups. The diversity in the magnitude of effects is important to take into account when considering the use of such information for personalized risk modeling.³⁰ Therefore caution should be used in extrapolating risks across different populations.

Our study provides strong and independent support for the role of 9 loci in the pathogenesis of PD in different populations. The detection and documentation of heterogeneity across different populations is useful in understanding the genetic architecture of this complex disease and in properly framing our ability to use this information in different clinical populations.

AUTHOR AFFILIATIONS

From the Department for Neurodegenerative Diseases (M.S., T.G., R.K.), Hertie Institute for Clinical Brain Research, University of Tübingen, and DZNE, German Centre for Neurodegenerative Diseases, Tübingen, Germany; Stanford Prevention Research Center (J.P.A.I.), Department of Medicine and Department of Health Research and Policy, Stanford University School of Medicine, Stanford, CA; Department of Neurology (J.O.A.), St Olavs Hospital and NTNU Trondheim, Trond-

heim, Norway; Institute of Neurological Sciences (G. Annesi, A.Q.), National Research Council, Mangone (CS), Italy; INSERM, UMR_S975 (A.B., S.L.), Université Pierre et Marie Curie-Paris, CNRS, UMR 7225, AP-HP, Pitié-Salpêtrière Hospital, Paris, France; Department of Genetics and Cytogenetics (A.B.), Pitié-Salpêtrière Hospital, Paris, France; Neurodegenerative Brain Diseases Group (C.V.B., D.C., J.T.), Department of Molecular Genetics, VIB, Antwerp; Institute Born-Bunge (C.V.B., D.C., J.T.), University of Antwerp, Antwerp, Belgium; Neuropsychiatric Genetics Group (L.B., C.M.L.), Department of Vertebrate Genomics, Max Planck Institute for Molecular Genetics, Berlin, Germany; General Hospital of Syros (M.B.), Syros; 10. "Hygeia" Hospital (M.B.), Clinic of Neurodegenerative Disorders, Athens; 2nd Neurology Clinic (M.B., L.S.), University of Athens, "Attikon" Hospital, Athens, Greece; Department of Neurology (D.C.), Antwerp University Hospital, Antwerp, Belgium; School of Clinical and Experimental Medicine (C.C., K.E.M.), College of Medical and Dental Sciences, University of Birmingham, and City Hospital, Birmingham, UK; Centre for Biomedicine (M. Facheris, A.A.H., P.P.P., I.P.), EURAC Bolzano, Bolzano, Italy; Department of Medical Genetics (M. Farrer, C.V.-G.), University of British Columbia, Vancouver, Canada; Cyclotron Research Centre (G.G.), Department of Neurology, University of Liège, Liège, Belgium; Department of Neurology (S.G., G. Auburger), Goethe University, Frankfurt am Main, Germany; Department of Neurology (G.M.H., G.X.), Faculty of Medicine, University of Thessaly and Institute of Biomedical Research and Technology, CERETETH, Larissa, Greece; Department of Neurology (N.H., H.T.), Juntendo University School of Medicine, Tokyo, Japan; Department of Neurology (B.J., S.S.P.), Seoul National University Hospital, Seoul, South Korea; Department of Neurology (C.M.L.), Johannes Gutenberg University, Mainz, Germany; Department of Neurology (J.-J.L., K.-C.Y.), Chushang Show-Chwan Hospital, Nantou and Chung-Shan Medical University Hospital, Taichung, Taiwan; The Dublin Neurological Institute at the Mater Misericordiae University Hospital (T.L.), and Conway Institute, University College Dublin, Ireland; Helmholtz Zentrum München (P.L.), German Research Centre for Environmental Health (GmbH), Neuherberg, Germany; Movement Disorders Centre (A.E.L.), Toronto Western Hospital, University of Toronto, and Edmond J Safra Program Parkinson's Disease, Toronto, Canada; Clinical Neurology Research Centre (V.M.), Department of Medicine & Therapeutics, The Chinese University of Hong Kong, Prince of Wales Hospital, Shatin, New Territories, Hong Kong; Department of Neurology (B.J.-M., G.O.), Medical University of Silesia, Katowice, Poland; Eskitis Institute for Cell and Molecular Therapies (G.D.M.), Griffith University, Australia; Neurosciences Department (K.E.M., J.D.S.), Queen Elizabeth Hospital Birmingham, University Hospitals Birmingham NHS Foundation Trust, Birmingham, UK; Tanz Centre for Research in Neurodegenerative Diseases (E.R.), Department of Medicine, University of Toronto, Toronto, Canada; Departments of Neuroscience (O.A.R.) and Neurology (R.J.U., Z.W.), Mayo Clinic, Jacksonville, FL; Divisions of Basic Neurosciences & Cell Biology (L. Stefanis), Biomedical Research Foundation of Academy of Athens, Athens, Greece; Division of Neurology/Molecular Brain Science (W.S., T.T.), Kobe University Graduate School of Medicine, Kobe, Japan; University of Queensland Centre for Clinical Research (P.A.S.), Herston, Australia; Department of Neurology (E.-K.T., Y.Z.), Singapore General Hospital, National Neuroscience Institute, Singapore; Department of Medical Epidemiology and Biostatistics and Department of Clinical Neuroscience (K.W.), Karolinska Institutet, Stockholm, Sweden; and Department of Neurology (D.M.), NorthShore University Health System, Chicago, IL.

AUTHOR CONTRIBUTIONS

Manu Sharma, PhD: Dr. Sharma designed the study, performed the statistical analysis, and wrote the manuscript. John P.A. Ioannidis, MD: Dr. Ioannidis performed the statistical analysis and participated in writing of the manuscript. Jan O Aasly, MD: Dr. Aasly contributed samples and collected phenotypic data. Grazia Annesi, MD: Dr. Annesi contributed samples and collected phenotypic data. Alexis Brice, MD: Dr. Brice contributed samples and collected phenotypic data. Christine Van Broeckhoven, PhD: Dr. Van Broeckhoven contributed samples and collected phenotypic data. Lars Bertram, MD: Dr. Bertram participated in writing the manuscript. Maria Bozi, MD: Dr. Bozi contributed samples and col-

lected phenotypic data. David Crosiers, MD: Dr. Crosiers contributed samples and collected phenotypic data. Carl Clarke, MD: Dr. Clarke contributed samples and collected phenotypic data. Maurizio Facheris, MD: Dr. Facheris contributed samples and collected phenotypic data. Matthew Farrer, PhD: Dr. Farrer contributed samples and collected phenotypic data. Gaetan Garraux, MD, PhD: Dr. Garraux contributed samples and collected phenotypic data. Suzana Gispert, PhD: Dr. Gispert contributed samples and collected phenotypic data. Georg Auburger, MD: Dr. Auburger contributed samples and collected phenotypic data. Carles Vilaríño-Güell, PhD: Dr. Vilaríño-Güell contributed samples and collected phenotypic data. Georgios M. Hadjigeorgiou, MD: Dr. Hadjigeorgiou contributed samples and collected phenotypic data. Andrew Hicks, PhD: Dr. Hicks contributed samples and collected phenotypic data. Nobutaka Hattori, MD: Dr. Hattori contributed samples and collected phenotypic data. Beom Jeon, MD: Dr. Jeon contributed samples and collected phenotypic data. Suzanne Lesage, MD: Dr. Lesage contributed samples and collected phenotypic data. Christina M Lill, MD: Dr. Lill participated in writing the manuscript. Juei-Jueng Lin, MD: Dr. Lin contributed samples and collected phenotypic data. Timothy Lynch, MD: Dr. Lynch contributed samples and collected phenotypic data. Peter Lichtner, PhD: Dr. Lichtner contributed samples and collected phenotypic data. Anthony E. Lang, MD: Dr. Lang contributed samples and collected phenotypic data. Vincent Mok, MD: Dr. Mok contributed samples and collected phenotypic data. Barbara Jasinska-Myga, MD, PhD: Dr. Jasinska-Myga contributed samples and collected phenotypic data. George D. Mellick, PhD: Dr. Mellick contributed samples and collected phenotypic data and critically reviewed the manuscript. Karen E Morrison, MD: Dr. Morrison contributed samples and collected phenotypic data. Grzegorz Opala, MD: Dr. Opala contributed samples and collected phenotypic data. Peter P. Pramstaller, PhD: Dr. Pramstaller contributed samples and collected phenotypic data. Irene Pichler, MD: Dr. Pichler contributed samples and collected phenotypic data. Sung Sup Park, MD: Dr. Park contributed samples and collected phenotypic data. Aldo Quattrone, MD: Dr. Quattrone contributed samples and collected phenotypic data. Ekaterina Rogaeva, PhD: Dr. Rogaeva contributed samples and collected phenotypic data and participated in writing of the manuscript. Owen A Ross, PhD: Dr. Ross contributed samples and collected phenotypic data. Leonidas Stefanis, MD: Dr. Stefanis contributed samples and collected phenotypic data. Joanne D. Stockton, PhD: Dr. Stockton contributed samples and collected phenotypic data. Wataru Satake, MD, PhD: Dr. Satake contributed samples and collected phenotypic data. Peter A. Silburn, PhD: Dr. Silburn contributed samples and collected phenotypic data. Jessie Theuns, PhD: Dr. Theuns contributed samples and collected phenotypic data. Eng-King Tan, MD: Dr. Tan contributed samples and collected phenotypic data. Tatsushi Toda, MD, PhD: Dr. Toda contributed samples and collected phenotypic data. Dr. Hiroyuki Tomiyama, MD, PhD: Dr. Tomiyama contributed samples and collected phenotypic data. Ryan J Uitti, MD: Dr. Uitti contributed samples and collected phenotypic data. Karin Wirdefeldt, MD: Dr. Wirdefeldt contributed samples and collected phenotypic data. Zbigniew Wszolek, MD: Dr. Wszolek contributed samples and collected phenotypic data. Georgios Xiromerisiou, MD: Dr. Xiromerisiou contributed samples and collected phenotypic data. Kuo-Chu Yueh, MD: Dr. Yueh contributed samples and collected phenotypic data. Yi Zhao, MD: Dr. Zhao contributed samples and collected phenotypic data. Thomas Gasser, MD: Dr. Gasser contributed samples and collected phenotypic data and participated in writing of the manuscript. Demetrius M. Maraganore, MD: Dr. Maraganore contributed samples and collected phenotypic data. Rejko Krüger, MD: Dr. Krüger contributed samples and collected phenotypic data and participated in writing of the manuscript.

DISCLOSURE

M. Sharma received funding support from Michael J Fox Foundation USA and travel grant from Movement Disorders Society. J.P.A. Ioannidis reports no disclosures. J.O. Aasly is supported by Norwegian Research Council and Reberg's Legacy. G. Annesi reports no disclosures. A. Brice receives grant support from Eranet Neuron (Co-PI), ANR Maladies Neurologiques et Psychiatriques 2008 (Co-PI), and PHRC National 2006 and 2008. C. Van Broeckhoven received funding from Research Foundation Flanders (FWO), research grant G.0085.08, co-PI; Research Foun-

dation Flanders (FWO), Belgium, postdoctoral fellowship, PI; Agency for Innovation by Science and Technology-Flanders (IWT), research grant 080020, co-PI; Flemish Government, Belgium, Methusalem excellence grant, key personnel; Belgian Science Policy Office, Interuniversity Attraction Poles program (IUAP) P6/43, key personnel; University of Antwerp, Belgium, Special Research Fund, key personnel; Foundation for Alzheimer Research (SAO-FRMA), Belgium; research grant 09007, PI; Alzheimer's Association, USA, New Investigator Research Grant, NIRG-09-132000, PI; Belgian Parkinson Foundation, Belgium, research grant, PI. L. Bertram, M. Bozi, D. Crosiers, C. Clarke, and M. Facheris report no disclosures. M. Farrer and Mayo Foundation received royalties from H. Lundbeck A/S and Isis Pharmaceuticals. In addition, Dr. Farrer has received an honorarium for a seminar at Genzyme. G. Garraux, S. Gispert, G. Auburger, C. Vilarinho-Güell, G.M. Hadjigeorgiou, and A. Hicks report no disclosures. N. Hattori has been serving as an advisory board member for Boehringer Ingelheim; as a result of attending advisory board meetings he received personal compensation. Nobutaka Hattori also has been serving as an advisory board member for FP Pharmaceutical Company by attending these advisory meetings he received personal compensation. He has been consulting with Ohtsuka Pharmaceutical Company, Kyowa Hakko Kirin Pharmaceutical Company, GlaxoSmithKline, Novartis, and Schering-Plough, and when he attended these advisory board meetings, he received personal compensation. B. Jeon has received funding for travel from GlaxoSmithKline Korea and Novartis Korea and has received research support as PI from Novartis, Boehringer Ingelheim, Ipsen Korea, the Korea Health 21 R&D Project, Ministry of Health & Welfare, Republic of Korea (#A030001), ABRC (Advanced Biometric Research Center), KOSEF (Korea Science and Engineering Foundation), Seoul National University Hospital, the Mr. Chung Suk-Gyoo and Sinyang Cultural Foundation, and the Song Foundation. S. Lesage, C.M. Lill, and J.-J. Lin report no disclosures. T. Lynch has served on an advisory board for Biogen and Novartis and has received honoraria from Lundbeck, Biogen, and Boehringer-Ingelheim. P. Lichtner reports no disclosures. A.E. Lang has served as an advisor for Abbott, Allon Therapeutics, Astra Zeneca, Biovail, Boehringer-Ingelheim, Cephalon, Ceregene, Eisai, Medtronic, Lundbeck A/S, Novartis, Merck Serono, Solvay, and Teva and received grants from Canadian Institutes of Health Research, Dystonia Medical Research Foundation, Michael J. Fox Foundation, National Parkinson Foundation, and Ontario Problem Gambling Research Centre and has served as an expert witness in cases related to the welding industry. V. Mok reports no disclosures. B. Jasinska-Myga was partially supported by the Robert and Clarice Smith Fellowship Program and the Pacific Alzheimer Research Foundation (PARF) C06-01 grant. G.D. Mellick reports no disclosures. K. E. Morrison has received grant support from Parkinson's UK, The Medical Research Council UK, the Wellcome Trust, and the Midlands Neurological Teaching and Research Fund. G. Opala was partially supported by the Robert and Clarice Smith Fellowship Program and the Pacific Alzheimer Research Foundation (PARF) C06-01 grant. P.P. Pramstaller, I. Pichler, S.S. Park, A. Quattrone, and E. Rogaeva report no disclosures. O.A. Ross is funded by NIH/NINDS awards P50 NS072187 and R01 NS078086, and the Michael J Fox Foundation. L. Stefanis has received a grant for genetic studies of PD from the Hellenic Secretariat of Research and Technology (PENED 2003), in which Novartis Hellas acted as a co-sponsor. J.D. Stockton reports no disclosures. W. Satake receives research support from Japan Intractable Disease Research Foundation. P.A. Silburn and J. Theuns report no disclosures. E.-K. Tan is funded by National Medical Research Council, Duke-NUS Graduate Medical School, and Singapore Millennium Foundation. He has received honoraria from Novartis, Boehringer Ingelheim, and GSK. T. Toda receives research support from Ministry of Health, Labour and Welfare of Japan. H. Tomiyama reports no disclosures. R.J. Uitti has received research funding from the NIH, PARRF, PSG, Noscira, Inc. and Advanced Neuromodulation Systems, Inc. Dr. Uitti has served as a Continuing Medical Educator for the AAN. His institution has received annual royalties from the licensing of the technology related to PARK8/LRRK2 greater than the federal threshold for significant financial interest. Dr. Uitti has not received any royalties. Dr. Uitti receives an honorarium as Associate Editor of *Neurology*[®]. K. Wierfeldt reports no disclosures. Z. Wszolek holds and has contractual rights for receipt of future royalty payments from patents re: A novel polynucle-

otide involved in heritable Parkinson's disease, receives educational research support from Allergan, Inc., the NIH/NINDS 1RC2NS070276, NS057567, P50NS072187, Mayo Clinic Florida (MCF) Research Committee CR programs (MCF #90052018 and MCF #90052030), and the gift from Carl Edward Bolch, Jr. and Susan Bass Bolch (MCF #90052031/PAU #90052). G. Xiomerisou, K.-C. Yueh, and Y. Zhao report no disclosures. T. Gasser is funded by Novartis Pharma, the Federal Ministry of Education and Research (BMBF) (NGFN-Plus and ERA-Net NEURON), the Helmholtz Association (HelMA, Helmholtz Alliance for Health in an Ageing Society), and the European Union (MeFoPa, Mendelian Forms of Parkinsonism). He received speaker honoraria from Novartis, Merck-Serono, Schwarz Pharma, Boehringer Ingelheim, and Valeant Pharma and royalties for his consulting activities from Cephalon Pharma and Merck-Serono. Dr. Gasser holds a patent concerning the LRRK2 gene and neurodegenerative disorders. D.M. Maraganore received funding support from the National Institutes of Health grant ES10751 ("Molecular Epidemiology of Parkinson's Disease"). He also received funding support from Alnylam Pharmaceuticals and Medtronic, Inc. to conduct an observational study of Parkinson's disease outcomes. Dr. Maraganore had a patent filed for a method to treat Parkinson disease. It has been licensed to Alnylam Pharmaceuticals. R. Krüger has received research grants of the German Research Council (DFG; KR2119/3-2), the Michael J Fox Foundation, and the Federal Ministry for Education and Research (BMBF, NGFNplus; 01GS08134), as well as speaker honoraria and/or travel grants from UCB Pharma, Cephalon, Abbott Pharmaceuticals, Takeda Pharmaceuticals, and Medtronic. **Go to Neurology.org for full disclosures.**

Received July 13, 2011. Accepted in final form January 25, 2012.

REFERENCES

- Manolio TA, Collins FS, Cox NJ, et al. Finding the missing heritability of complex diseases. *Nature* 2009;461:747–753.
- McCarthy MI, Abecasis GR, Cardon LR, et al. Genome-wide association studies for complex traits: consensus, uncertainty and challenges. *Nat Rev* 2008;9:356–369.
- Simon-Sanchez J, Schulte C, Bras JM, et al. Genome-wide association study reveals genetic risk underlying Parkinson's disease. *Nat Genet* 2009;41:1308–1312.
- Maraganore DM, de Andrade M, Lesnick TG, et al. High-resolution whole-genome association study of Parkinson disease. *Am J Hum Genet* 2005;77:685–693.
- Pankratz N, Wilk JB, Latourelle JC, et al. Genomewide association study for susceptibility genes contributing to familial Parkinson disease. *Hum Genet* 2009;124:593–605.
- Satake W, Nakabayashi Y, Mizuta I, et al. Genome-wide association study identifies common variants at four loci as genetic risk factors for Parkinson's disease. *Nat Genet* 2009;41:1303–1307.
- Saad M, Lesage S, Saint-Pierre A, et al. Genome-wide association study confirms BST1 and suggests a locus on 12q24 as the risk loci for Parkinson's disease in the European population. *Hum Molecul Genet* 20:615–627.
- Spencer CC, Plagnol V, Strange A, et al. Dissection of the genetics of Parkinson's disease identifies an additional association 5' of SNCA and multiple associated haplotypes at 17q21. *Hum Molecul Genet* 20:345–353.
- Hamza TH, Zabetian CP, Tenesa A, et al. Common genetic variation in the HLA region is associated with late-onset sporadic Parkinson's disease. *Nat Genet* 42:781–785.
- Maraganore DM, de Andrade M, Elbaz A, et al. Collaborative analysis of alpha-synuclein gene promoter variability and Parkinson disease. *JAMA* 2006;296:661–670.

11. Simon-Sanchez J, van Hilten JJ, van de Warrenburg B, et al. Genome-wide association study confirms extant PD risk loci among the Dutch. *Eur J Hum Genet* 2011;19: 655–661.
12. Mata IF, Yearout D, Alvarez V, et al. Replication of MAPT and SNCA, but not PARK16-18, as susceptibility genes for Parkinson's disease. *Mov Disord* 2011;26:819–823.
13. Simon-Sanchez J, van Hilten JJ, van de Warrenburg B, et al. Genome-wide association study confirms extant PD risk loci among the Dutch. *Eur J Hum Genet* 2011;19: 655–661.
14. Nalls MA, Plagnol V, Hernandez DG, et al. Imputation of sequence variants for identification of genetic risks for Parkinson's disease: a meta-analysis of genome-wide association studies. *Lancet* 2011;377:641–649.
15. Ioannidis JP, Thomas G, Daly MJ. Validating, augmenting and refining genome-wide association signals. *Nat Rev* 2009;10:318–329.
16. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Stat Med* 2002;21:1539–1558.
17. Ioannidis JP, Patsopoulos NA, Evangelou E. Uncertainty in heterogeneity estimates in meta-analyses. *BMJ* 2007; 335:914–916.
18. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ* 2003;327: 557–560.
19. Clarke GM, Cardon LR. Aspects of observing and claiming allele flips in association studies. *Genet Epidemiol* 2010;34:266–274.
20. Rhodes SL, Sinsheimer JS, Bordelon Y, Bronstein JM, Ritz B. Replication of GWAS associations for GAK and MAPT in Parkinson's disease. *Ann Hum Genet* 2011;75:195–200.
21. Do CB, Tung JY, Dorfman E, et al. Web-based genome-wide association study identifies two novel Loci and a substantial genetic component for Parkinson's disease. *PLoS Genet* 2011;7:e1002141.
22. Nalls MA, Plagnol V, Hernandez DG, et al. Imputation of sequence variants for identification of genetic risks for Parkinson's disease: a meta-analysis of genome-wide association studies. *Lancet* 2011;377:641–649.
23. Ioannidis JP, Patsopoulos NA, Evangelou E. Heterogeneity in meta-analyses of genome-wide association investigations. *PLoS One* 2007;2:e841.
24. Thompson SG. Why sources of heterogeneity in meta-analysis should be investigated. *BMJ* 1994;309:1351–1355.
25. Lewis SJ, Foltynie T, Blackwell AD, Robbins TW, Owen AM, Barker RA. Heterogeneity of Parkinson's disease in the early clinical stages using a data driven approach. *J Neurol Neurosurg Psychiatry* 2005;76:343–348.
26. Wang K, Li M, Hakonarson H. Analysing biological pathways in genome-wide association studies. *Nat Rev Genet* 2010;11:843–854.
27. Fuchs J, Nilsson C, Kachergus J, et al. Phenotypic variation in a large Swedish pedigree due to SNCA duplication and triplication. *Neurology* 2007;68:916–922.
28. Scholz SW, Houlden H, Schulte C, et al. SNCA variants are associated with increased risk for multiple system atrophy. *Ann Neurol* 2009;65:610–614.
29. Zody MC, Jiang Z, Fung HC, et al. Evolutionary toggling of the MAPT 17q21.31 inversion region. *Nat Genet* 2008; 40:1076–1083.
30. Ioannidis JP, Ntzani EE, Trikalinos TA. 'Racial' differences in genetic effects for complex diseases. *Nat Genet* 2004;36:1312–1318.

Visit the *Neurology*[®] Web Site at www.neurology.org

- Enhanced navigation format
- Increased search capability
- Highlighted articles
- Detailed podcast descriptions
- RSS Feeds of current issue and podcasts
- Personal folders for articles and searches
- Mobile device download link
- AAN Web page links
- Links to Neurology Now[®], Neurology Today[®], and Continuum[®]
- Resident & Fellow subsite

 Find *Neurology*[®] on Facebook: <http://tinyurl.com/neurologyfan>

 Follow *Neurology*[®] on Twitter: <http://twitter.com/GreenJournal>